

Integrating using inverse hyperbolic trig

Key Points

By considering the derivatives of inverses we get integrations to do with the reciprocal of squares.

The integrals can be found in the formula book. Including logarithm forms

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \cosh^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \sinh^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{a} \tanh^{-1}\left(\frac{x}{a}\right)$$

Basic Skills

Q1. Write the logarithm form of

a) $\operatorname{arctanh} x$

b) $\operatorname{arcsinh} x$

c) $\operatorname{arccosh} x$

Q2. Solve in logarithmic form $\cosh x = 5$

Q3. For $y = \sinh^{-1} x$

a) State $\frac{dy}{dx}$ using the formula in the key point section/formula book

b) By using $x = \sinh y$ prove your statement in part a)

Competency Skills

Q4. Find the following integrals

a) $\int \frac{1}{\sqrt{x^2 + 25}} dx$ b) $\int \frac{1}{\sqrt{4x^2 - 200}} dx$ c) $\int \frac{1}{x^2 - 30} dx$

Q5. Evaluate the following giving your answer in logarithm form

a) $\int_3^5 \frac{1}{\sqrt{3x^2 - 9}} dx$ b) $\int_0^1 \frac{5}{\sqrt{25x^2 + 1}} dx$ c) $\int_0^2 \frac{2}{20 - 2x^2} dx$

Q6. Find the integral of $\frac{1}{\sqrt{(3x-4)^2 - 25}}$

Q7. Given $\int \frac{1}{1-x^2} dx = \tanh^{-1}(x)$ explain why $\int \frac{1}{a^2 - x^2} dx = \frac{1}{a} \tanh^{-1}\left(\frac{x}{a}\right)$

Advanced Skills

Q8. Evaluate the integrals

a) $\int_1^2 \frac{2}{\sqrt{x^2 - 4x + 5}} dx$ b) $\int \frac{1}{\sqrt{2x^2 + 4x - 16}} dx$ c) $\int \frac{1}{20 - 6x - 2x^2} dx$

Q9. Prove $\int \frac{1}{\sqrt{x^2 + 1}} dx = \sinh^{-1}(x)$ using the substitution that $x = \sinh u$

Q10. Prove $\int \frac{1}{1-x^2} dx = \tanh^{-1}(x)$ using the substitution that $x = \tanh u$

Q11. Find $\int \sqrt{x^2 - 1} dx$ by using an appropriate substitution

Extension

Q12. Explain the difference between the use of inverse standard trig and inverse hyperbolic trig in what they can achieve using their derivatives for integration